Bang-time perturbations and redshift drift in a parabolic LTB model: a reproducible light-cone calculation

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Abstract

We calculate the response of redshift drift to a radially varying bang time in an exactly solvable, parabolic Lemaître–Tolman–Bondi (LTB) dust spacetime. The drift is propagated between neighboring light cones rather than inferred from a Friedmann–Lemaître–Robertson–Walker (FLRW) endpoint formula. We derive an explicit variational system and compare its response with finite differences. Two smooth, centrally regular bang-time perturbations yield negative equal-luminosity-distance responses at all 15 sampled baseline redshifts $0.1 \le z \le 1.5$; no zero crossing appears in these examples. We quantify how equal-distance and fixed-redshift comparisons differ, and how the FLRW endpoint shortcut misestimates the response. This dimensionless dust example is neither an observational fit nor a uniqueness result. We identify the functional distance compensation, background realism, and physical constraints needed for a cosmological inference.

1 Scope and relation to earlier work

Redshift drift in spherically symmetric spacetimes and its use with distance data were established by Uzan, Clarkson and Ellis [1]. Yoo, Kai and Nakao developed the LTB drift calculation and studied void models [2]. Exact growing and decaying density modes in LTB dust have been analyzed by Sussman [3]. The inverse problem for LTB observables has additional existence and regularity conditions [4]. We therefore claim neither a new general drift equation nor a unique observational signature. The purpose here is to correct an endpoint-formula error in an earlier unpublished draft, spell out an explicit parameter derivative, and give a small reproducible test case.

In units $c = G = 1$, the LTB metric and evolution equation are

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \frac{R'(t,r)^2}{1+2E(r)}\mathrm{d}r^2 + R(t,r)^2\mathrm{d}\Omega^2, \tag{1}$$

$$\dot{R}^2 = \frac{2M(r)}{R} + 2E(r), \qquad R(t_{\mathrm{B}}(r), r) = 0. \tag{2}$$

Dots and primes denote partial derivatives at fixed r and t, respectively. A radial coordinate choice removes one of M, E, t_{B} as an independent function. A nonconstant t_{B} represents the decaying mode in the conventional LTB dust mode decomposition [3]; its additive constant sets the time origin. Throughout this paper the mass-coordinate gauge and the profiles below are specified explicitly.

The calculation is organized around five questions. (i) What is the drift of the *same comoving source* when the central observer's clock advances? (ii) Does a bang-time perturbation give the same derivative at fixed redshift and at equal luminosity distance? (iii) Does either of the specified perturbations force a zero crossing? (iv) How large is the error from importing the FLRW endpoint expression into

LTB? (v) What further constraints are required before a response can be identified as an observable decaying-mode signal? Equations (3)–(9) answer the first two mathematically; Secs. 4–6 test the middle questions and delimit the last one.

2 Drift on neighboring radial light cones

Put $S(r) = \sqrt{1+2E(r)}$, $A = R'/S$, $B = \dot{R}'/S$, and $C = \ddot{R}'/S$. On an incoming radial ray from a central, comoving observer at $r = 0$, the emission time $t(r)$ and observed redshift $z(r)$ satisfy

$$\frac{\mathrm{d}t}{\mathrm{d}r} = -A, \qquad \frac{\mathrm{d}z}{\mathrm{d}r} = (1+z)B, \qquad t(0) = t_0, \quad z(0) = 0. \tag{3}$$

Compare two rays received at times separated by δt_0, from the *same comoving emitter* r. Their time separation $T(r) = \partial t(r; t_0)/\partial t_0$ obeys $T' = -BT$, $T(0) = 1$, hence $T = 1/(1+z)$. Let $Z(r) = \partial z(r; t_0)/\partial t_0$ be drift per unit central proper time. Differentiating Eq. (3) with respect to t_0 at fixed r gives the exact propagation equation

$$Z' = BZ + (1+z)CT = BZ + C, \qquad Z(0) = 0, \tag{4}$$

or, equivalently,

$$\boxed{Z(r) = (1+z(r)) \int_0^r \frac{C(\bar{r})}{1+z(\bar{r})}\,\mathrm{d}\bar{r}.} \tag{5}$$

This result carries the information along the ray. In a homogeneous FLRW limit it reduces to $Z = (1+z)H_0 - H(t_e)$. In a general LTB geometry, replacing Eq. (4) by $(1+z)H_0 - \dot{R}'/R'|_e$ is not justified: the gradient of expansion along the ray contributes. For an isotropic central observer, reciprocity gives

$$D_L(r) = (1+z(r))^2 R(t(r), r). \tag{6}$$

3 Linear response and what "fixed distance" means

Let p denote a model parameter and $\boldsymbol{y} = (t, z, Z)$. With $\boldsymbol{y}' = \boldsymbol{F}(\boldsymbol{y}, r, p)$ from Eqs. (3) and (4), its first-order variation $\boldsymbol{u} = \partial_p \boldsymbol{y}|_r$ is specified, without finite differences, by

$$\boldsymbol{u}' = \underbrace{\begin{pmatrix} -A_t & 0 & 0 \\ (1+z)B_t & B & 0 \\ B_t Z + C_t & 0 & B \end{pmatrix}}_{\boldsymbol{F}_{\boldsymbol{y}}} \boldsymbol{u} + \begin{pmatrix} -A_p \\ (1+z)B_p \\ B_p Z + C_p \end{pmatrix}, \qquad \boldsymbol{u}(0) = \boldsymbol{0}. \tag{7}$$

All partial derivatives in this equation are at fixed t, r; in particular, A_p includes any parameter dependence in S. Equation (7) is the analytic variational equation used to define the linear response; the public numerical example evaluates its derivative by symmetric finite differences and checks step convergence.

At a given emitter coordinate the distance response is

$$\partial_p D_L|_r = (1+z)^2(R_t t_p + R_p) + 2(1+z)R z_p, \qquad \frac{\mathrm{d}D_L}{\mathrm{d}r} = 2(1+z)z_r R + (1+z)^2(R_t t_r + R_r). \tag{8}$$

Here R_p and R_r are partial derivatives, whereas $\mathrm{d}D_L/\mathrm{d}r$ is a total derivative along the ray. If $\mathrm{d}D_L/\mathrm{d}r \neq 0$, implicit differentiation yields the response at a *single equal luminosity distance*,

$$\boxed{\left.\frac{\partial Z}{\partial p}\right|_{D_L} = Z_p|_r - \frac{Z_r}{\mathrm{d}D_L/\mathrm{d}r}\,\partial_p D_L|_r.} \tag{9}$$

Table 1: Diagnostics produced by `python analysis.py`. Extremes and convergence refer to the 15 sampled distances, not a continuum bound.

w	$\min \partial_\epsilon Z\|_{D_L}$	$\max \partial_\epsilon Z\|_{D_L}$	max. change on halving h
0.8	-0.328043	-0.001464	1.39×10^{-7}
1.2	-1.282333	-0.001490	1.81×10^{-7}

For comparison, $\partial_p Z|_z = Z_p|_r - (Z_r/z_r)z_p|_r$. Neither condition requires $D_L(z)$ to be identical between the two models. Keeping the *whole function* $D_L(z)$ fixed would require a compensating functional perturbation in M or E and is a distinct inverse problem. Equations (7)–(9) also show that a zero-crossing location cannot be inferred from the background profile alone without specifying the allowed perturbation and compensation.

4 Exactly specified illustrative model

Set $t_0 = 1$, $E(r) = 0$, $M(r) = 2r^3/9$, and $t_{\rm B}(r) = \epsilon f_w(r)$, with

$$f_w(r) = r^2 \exp[-(r/w)^2], \qquad w \in \{0.8, 1.2\}. \tag{10}$$

All coordinates and drift values in this section are dimensionless. The two choices of w are perturbation shapes on the *same homogeneous background*, not two realistic void profiles. The even behavior at $r = 0$ makes the center smooth. The exact expanding solution, with $\tau = t - \epsilon f_w(r)$, is

$$\begin{aligned} R &= r\tau^{2/3}, \\ R' &= \tau^{2/3} - \frac{2r\epsilon f_w'}{3\tau^{1/3}}, \quad \dot{R}' = \frac{2}{3\tau^{1/3}} + \frac{2r\epsilon f_w'}{9\tau^{4/3}}, \\ \ddot{R}' &= -\frac{2}{9\tau^{4/3}} - \frac{8r\epsilon f_w'}{27\tau^{7/3}}. \end{aligned} \tag{11}$$

The expansion branch requires $\tau > 0$; the code checks $R' > 0$ and monotonic z, D_L on every sampled light cone. At $\epsilon = 0$, the model is Einstein–de Sitter (EdS) and has the exact check

$$Z_{\rm EdS}(z) = \frac{2}{3}\big[(1+z) - (1+z)^{3/2}\big]. \tag{12}$$

At $\epsilon \neq 0$ we hold M, E, t_0 fixed, integrate to $z = 1.6$, and compare drift values at the *baseline* $D_L(z)$ for $z = 0.1, 0.2, \ldots, 1.5$. The derivative uses $[Z(+h) - Z(-h)]/(2h)$ at $h = 0.002, 0.001, 0.0005$. The plotted result uses $h = 0.0005$.

At $\epsilon = 0$, the full null-cone solution provides further independent checks:

$$t(r) = (1 - r/3)^3, \qquad 1 + z(r) = (1 - r/3)^{-2}, \qquad D_L(z) = 3\big[(1+z) - \sqrt{1+z}\big]. \tag{13}$$

The source radius corresponding to a baseline redshift is $r(z) = 3[1 - (1+z)^{-1/2}]$. It is this baseline distance and radius, rather than the perturbed model's redshift label, that define the equal-distance sample.

The maximum absolute baseline error against Eq. (12) on this grid is 4.4×10^{-13}. An independent two-observer-time finite difference of redshift at fixed $r = 0.8$, with $\epsilon = 0.001$ and time step 10^{-5}, agrees with the propagated drift to 3.3×10^{-11}. For $\epsilon = 0.001$, the first-order maximum fractional distance change at fixed baseline z is approximately 2.20×10^{-4} ($w = 0.8$) or 3.78×10^{-4} ($w = 1.2$). Thus the equal-distance operation is numerically distinct from holding the complete distance relation fixed.

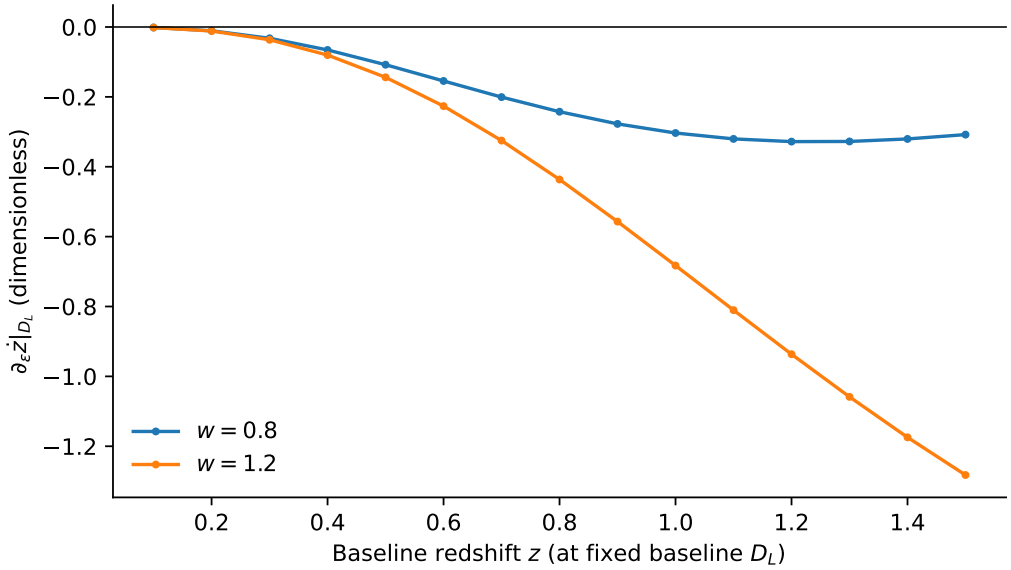


Figure 1: Dimensionless bang-time response at equal D_L for the two specified f_w. The horizontal coordinate labels the *baseline* redshift. Lines guide the eye between 15 sampled points; no observational uncertainties are implied.

Table 2: Response derivatives at baseline $z = 1$ (dimensionless). The last column applies the FLRW endpoint shortcut at fixed z and is not a valid LTB drift calculation.

w	exact, fixed z	exact, equal D_L	endpoint shortcut, fixed z
0.8	-0.158830	-0.303455	$+0.164275$
1.2	-0.426856	-0.682806	-2.670339

Table 1 and Fig. 1 support the narrow statement that these two examples have no sign change at the sampled distances. They do not establish a sign theorem for arbitrary f, nonparabolic backgrounds, or compensated models.

The linear variational calculation in Appendix A agrees with the $h = 0.0005$ equal-distance finite difference to at most 4.7×10^{-8} for $w = 0.8$ and 6.1×10^{-8} for $w = 1.2$ on the grid. Figure 2 and Table 2 make the comparison choice and endpoint-formula error visible rather than hiding them inside a single response curve.

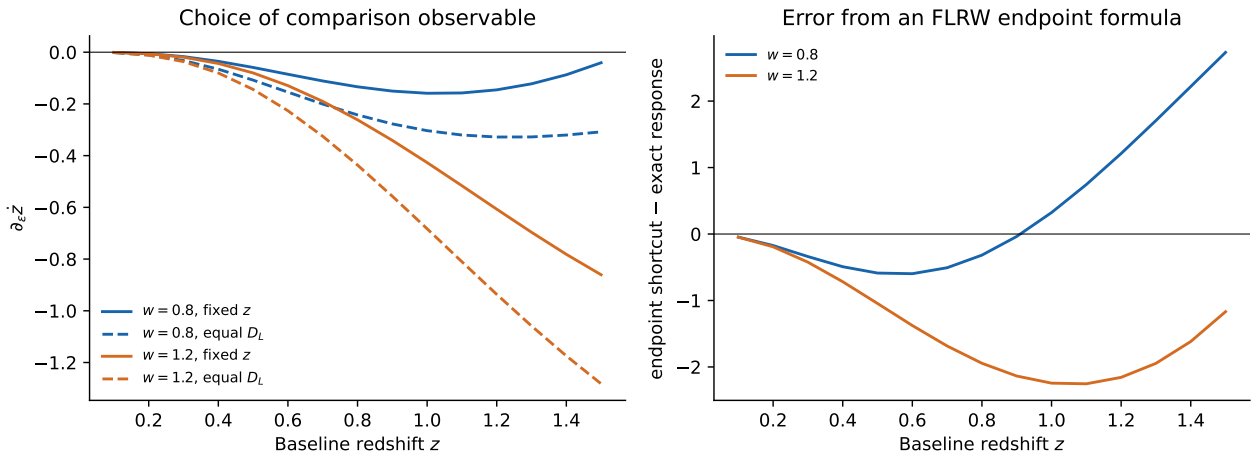


Figure 2: *Left:* Fixed-z (solid) and equal-D_L (dashed) derivatives for each bang-time width. *Right:* Derivative from the FLRW endpoint shortcut minus the correct fixed-z drift derivative. This difference can change sign with perturbation shape. Both panels use the same dimensionless parameter and baseline redshift grid.

5 Questions left open

1. *Functional distance compensation.* Can $\delta M(r)$ or $\delta E(r)$ be chosen so that $\delta D_L(z) = 0$ over a continuous redshift interval while retaining a nonzero $\delta Z(z)$? This requires solving the linearized inverse problem with a radial gauge, boundary conditions, and regularity criteria. Equation (9) only compares at one distance at a time.

2. *Profile dependence.* Do any admissible, smooth bang-time profiles generate a robust zero crossing after such compensation? The two f_w examples give a negative answer only for their sampled, uncompensated family. A crossing cannot be claimed universal or excluded universally from these results.

3. *Physical backgrounds.* How do the response and its sign change in an inhomogeneous growing-mode void or an LTB model with Λ? A calibrated expansion history and physically allowed profiles are needed before translating the dimensionless derivative into a velocity shift.

4. *Viability and inference.* What bounds on ϵf_w follow from early-time physics, shell-crossing avoidance, and independent observables, and what drift precision remains after marginalizing other LTB functions? Positivity of R' on the sampled past ray alone is not a global regularity proof.

6 Interpretation and limitations

The present parabolic EdS background has no growing-mode void, dark energy, realistic expansion calibration, observational covariance, or early-Universe consistency test. Bang-time variations can generate strong early-time effects and are constrained by other observations; this manuscript does not estimate allowed amplitudes. The plots give derivatives in chosen dimensionless coordinates, not a predicted velocity drift in cm/s over a survey baseline. A physical forecast needs a calibrated model and nuisance effects, along with a functional distance compensation and a treatment of observational

uncertainties. This preprint therefore provides a corrected calculational baseline, not evidence for a detected or uniquely isolatable decaying mode.

Reproducibility

The accompanying `analysis.py`, dependency list, both vector figures and machine-readable numbers fix every numerical input. Execute `python analysis.py` followed by `pdflatex paper.tex` twice. The script checks positive R' and monotonic redshift and distance along each sampled ray, the EdS limit, the two-observer-time derivative and the linear variational result. No observed data or generated artwork enters the result.

A Explicit forcing of the bang-time variation

For the profile of Eq. (10), define $g(r) = f_w(r) + r f_w'(r)$. Holding t, r fixed and differentiating Eq. (11) at $\epsilon = 0$ gives

$$A_\epsilon = -\frac{2g}{3t^{1/3}}, \qquad B_\epsilon = \frac{2g}{9t^{4/3}}, \qquad C_\epsilon = -\frac{8g}{27t^{7/3}}. \tag{14}$$

The baseline time derivatives needed in Eq. (7) are

$$A_t = \frac{2}{3t^{1/3}}, \qquad B_t = -\frac{2}{9t^{4/3}}, \qquad C_t = \frac{8}{27t^{7/3}}. \tag{15}$$

We integrate Eq. (7) with these forcing terms along Eq. (13), then apply Eqs. (8) and (9). This analytically differentiated initial-value problem provides a check independent of changing ϵ in the nonlinear ray integrator. Agreement on the sampled interval tests the implementation; it does not guarantee convergence at arbitrarily high redshift or global absence of shell crossings.

References

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